

Technology #09

UWB Catheter Position Mapping with
SAR-Bounded Accuracy

Engineering Technology-Transfer Dossier
CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION
Generated: 2026-08-27

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

UWB Catheter Position Mapping with SAR-Bounded Accuracy

WHY DOES IT MATTER?

Catheter position verification requires CT/MRI (expensive, ionizing for CT)

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 5 governing equations, 12 design inputs, 8 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in Radio Frequency (UWB) Localization + Tissue Propagation. 7-step build plan with specific test articles and acceptance criteria. Target buyer: Medical device localization company; neuroshunt manufacturer.

WHAT DOES THE BUYER GET?

UWB localization concept + R1 correction (SAR-bounded accuracy); TOA + SAR + tissue propagation governing equations; Bench test protocols (link budget, SAR, accuracy, EMC, FCC); External precedent catalog (UWB medical radar, FCC Part 15.250, IEEE 1528); Critical UNKNOWN disclosure (achievable accuracy, antenna efficiency, multipath); This engineering dossier with failure analysis + build plan

WHAT DOES THE BUYER HAVE TO BUILD?

Production UWB transmitter design (IC selection, modulation, power management); Implantable UWB antenna design (small form factor); External receiver array (production hardware + localization software); Manufacturing process for catheter-integrated UWB implant; Regulatory submission (FDA + FCC combination; complex); Clinical validation (localization accuracy vs imaging); FCC certification + IEC 60601-1-2 compliance

WHAT IS THE NEXT DECISIVE EXPERIMENT?

UWB TX + RX link bench test

WHAT WOULD MAKE US KILL IT?

SAR limit (1.6 W/kg) constrains transmit power below useful SNR at >5cm tissue depth

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 5 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 7-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: UWB TX + RX link bench test

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - UWB TX + RX link bench test - 6 weeks

Basis: Build plan WP. Scope: Path loss vs distance in air + simple tissue phantom. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - SAR simulation model - 8 weeks

Basis: Build plan WP. Scope: SAR distribution at 1g averaging. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - SAR measurement in phantom - 6 weeks

Basis: Build plan WP. Scope: SAR at 1g averaging. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 5 equations. Key assumptions: Tissue approximated as layered dielectric (skin, fat, muscle, CSF) with published properties; Multipath separable given UWB bandwidth (≥ 500 MHz); External receiver array geometry known precisely; Implant clock synchronized to external reference (or differential TDOA used to relax requirement); FCC Part 15.250 applies (UWB medical imaging). UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

5 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: UWB TX + RX link bench test. Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: UWB transmitter in catheter; external receiver array localizes via TOA with SAR-bounded accuracy (R1 fix)

POTENTIAL VALUE DRIVER: Addresses: Catheter position verification requires CT/MRI (expensive, ionizing for CT)

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: Medical device localization company; neuroshunt manufacturer

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: Radio Frequency (UWB) Localization + Tissue Propagation

Disciplines: RF engineering (UWB transceiver design), Antenna engineering (implantable antennas), Tissue electromagnetics (dielectric properties, propagation), Regulatory engineering (FCC SAR, IEC 60601-1-2), Signal processing (TOA estimation, multipath mitigation)

Architecture: Implantable UWB transmitter in shunt catheter; external receiver array localizes catheter position via time-of-arrival (TOA) ranging. R1 fix: original P-21 assumed sub-mm accuracy without SAR analysis; R1 redesign bounds accuracy by SAR limit and tissue propagation physics.

Subsystems:

SS-01: Implantable UWB transmitter - Emits UWB pulse train at FCC Part 15.250 compliant frequency (3.1-10.6 GHz band, BPSK or pulse UWB)

SS-02: Implantable antenna (small-form-factor) - Radiates UWB signal through tissue; antenna geometry constrained by catheter diameter (~2 mm)

SS-03: External receiver array (4+ antennas) - Receives UWB signal at multiple known positions; computes TOA for localization

SS-04: Localization algorithm (TOA + TDOA) - Estimates 3D position of implant from TOA measurements at multiple receivers

2. Mechanism Architecture

Physical Changes: Implant emits UWB pulse; signal propagates through tissue with frequency-dependent attenuation; external array receives signal; TOA at each receiver yields position estimate

Key Physics: UWB pulse propagation through tissue: signal attenuates with frequency (higher freq more attenuated) and distance. Tissue dielectric properties (permittivity, conductivity) vary by tissue type. TOA-based ranging accuracy bounded by Cramer-Rao lower bound: $\sigma_{\text{TOA}} \geq c / (2\pi\beta\sqrt{2\text{SNR}})$ where β is RMS bandwidth.

3. Governing Engineering Model

Summary: UWB time-of-arrival ranging through tissue. Localization accuracy bounded by signal bandwidth, SNR (limited by SAR-constrained transmit power and tissue attenuation), and array geometry (GDOP).

Equations:

$d = c \cdot dt$ [P-21-R1 UWB TOA from implant to external receiver array]

$\text{SAR} = (\sigma \cdot E^2) / (2 \cdot \rho)$ [P-21-R1 SAR constraint on UWB implant transmit power]

$\sigma_{\text{TOA}} \geq c / (2\pi\beta\sqrt{2\text{SNR}})$ [Cramer-Rao lower bound]

$\sigma_{\text{pos}} = \text{GDOP} \cdot \sigma_{\text{TOA}}$ [geometric dilution of precision]

$\text{PL}_{\text{tissue}}(d, f) = \alpha(f) \cdot d$ [frequency-dependent path loss]

Assumptions:

- Tissue approximated as layered dielectric (skin, fat, muscle, CSF) with published properties
- Multipath separable given UWB bandwidth (≥ 500 MHz)
- External receiver array geometry known precisely
- Implant clock synchronized to external reference (or differential TDOA used to relax requirement)
- FCC Part 15.250 applies (UWB medical imaging)

Boundary Conditions:

- Implant depth: 1-10 cm typical (cranial to peritoneal)
- Tissue types: skin, fat, muscle, CSF, bone
- Frequency: 3.1-10.6 GHz UWB
- SAR limit: 1.6 W/kg averaged over 1g tissue

Failure Regimes:

- SAR limit reached before sufficient SNR -> accuracy insufficient
- Tissue attenuation > link budget -> no signal at receiver
- Multipath dominates -> TOA biased
- Array geometry poor ($\text{GDOP} > 5$) -> localization inaccurate
- Implant antenna inefficient at UWB in small form factor -> transmit power wasted

4. Design Inputs

ID	Input	Value	Origin
DI-001	Clinical need	Catheter position verification post-implantation and during follow-up; current standard: CT/MRI (expensive, ionizing for patient)	VERIFIED
DI-002	Functional requirement	UWB-based localization of catheter position; R1 fix: accuracy bounded by SAR and UWB physics	MODELLED
DI-003	Frequency band	EXTERNAL_PRECEDENT (UWB 3.1-10.6 GHz per FCC Part 15.250)	EXTERNAL_PRECEDENT
DI-004	SAR limit	EXTERNAL_PRECEDENT (1.6 W/kg averaged over 1g tissue, FCC)	EXTERNAL_PRECEDENT
DI-005	Tissue dielectric properties	EXTERNAL_PRECEDENT (Gabriel et al. published measurements)	EXTERNAL_PRECEDENT
DI-006	Bandwidth	EXTERNAL_PRECEDENT (>= 500 MHz per FCC UWB definition)	EXTERNAL_PRECEDENT
DI-007	Localization accuracy target	UNKNOWN — clinical utility threshold TBD	UNKNOWN
DI-008	Implant antenna size constraint	Constrained by catheter diameter ~2 mm	VERIFIED
DI-009	Biocompatibility (ISO 10993)	UNKNOWN	UNKNOWN
DI-010	Sterilization	UNKNOWN	UNKNOWN
DI-011	EMC	UNKNOWN — required IEC 60601-1-2	UNKNOWN
DI-012	Battery life (if active TX)	UNKNOWN	UNKNOWN

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	Implantable UWB transmitter design	ABSENT	Center frequency; Modulation scheme; Power budget; Antenna design
DO-002	Implantable antenna (small form factor UWB)	ABSENT	Geometry constraints; Center frequency; Tissue loading analysis
DO-003	External receiver array	CONCEPTUAL	
DO-004	Localization algorithm	MODELLED	

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
Center frequency f_c	UNKNOWN — design choice based on	Hz	Antenna resolution Design standard	Tissue phantom measurement
Bandwidth B	EXTERNAL_PRECEDENT (UWB requires >= 500 MHz)	Hz	FCC Part 15.250	
Transmit power P_tx	UNKNOWN — bounded by SAR limit	W	FCC SAR constraint	SAR simulation + measurement per IEEE 1528
SAR limit	EXTERNAL_PRECEDENT (1.6 W/kg averaged over 1g)	W/kg	FCC standard (FCC)	
Implant antenna size	UNKNOWN — constrained by catheter diameter	cm	Catheter geometry	Geometric verification with catheter vendor
Receiver array size	UNKNOWN	count	Localization accuracy target	
Tissue dielectric properties	EXTERNAL_PRECEDENT (published)	—	Published properties	

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
SAR limit too restrictive	SAR limit caps transmit power	Duty cycle reduction	SAF power, wider bandwidth	IEEE 1528 (Max SAR per pulse), Extra SAR power coupling
Tissue attenuation excessive	High frequency UWB attenuated by tissue	Lower frequency (less tissue loss)	Tissue phantom measurements	UNKNOWN optimal frequency
Multipath bias	Reflected signals bias TOA estimate	Multipath detection algorithm	Phantom with reflecting surfaces	UNKNOWN multipath magnitude in vivo
Poor array geometry (High GDOP)	High GDOP geometry amplifies TOA error	Optimal array placement + redundancy	GDOP estimators - phantom	UNKNOWN achievable GDOP in clinical setting
Implant antenna inefficient	Small antenna at UWB has low efficiency	Antenna optimization + matching	Antenna measurement in tissue	UNKNOWN achievable efficiency in 2 mm form factor
Clock drift between implant and external	Implant and external not synchronized	EDC (differential) instead of TOA	Localization testing	UNKNOWN clock stability requirement
EMC interference with UWB devices	UWB devices interfere with each other	FCC Part 15 compliance	IEC 60601-1-2 IEC 60601-1-2	UNKNOWN specific interferences
Battery depletion (implant)	Active UWB transmitter consumes power	Duty cycling + low-power design	Significance harvesting (BIS)	UNKNOWN achievable battery life

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
SAR limit too restrictive	SAR cap limits P_tx below useful SNR	Duty cycling + wider bandwidth + external antenna coupling	UNKNOWN achievable SNR
Tissue attenuation excess	High-frequency UWB attenuated	Lower frequency (tradeoff with resolution)	UNKNOWN optimal frequency
Multipath bias	Reflected signals bias TOA	First-path detection + UWB bandwidth	UNKNOWN in vivo multipath
Poor array geometry (high GDOP)	Geometry amplifies TOA error	Optimal placement + redundant receivers	UNKNOWN achievable GDOP clinically
Implant antenna inefficient	Small antenna low efficiency at UWB	Antenna optimization + matching network	UNKNOWN achievable efficiency in 2 mm
Clock drift	Implant clock not synchronized	TDOA or two-way ranging	UNKNOWN clock stability requirement
EMC interference	UWB emissions affect other devices	FCC + IEC 60601-1-2 compliance	UNKNOWN specific interferences
Battery depletion	Active UWB TX consumes power	Duty cycling + low-power design + energy harvesting	UNKNOWN battery life

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
V-001	Localization accuracy in tissue phantom	Phantom + UWB localization	$\sigma_{\text{pos}} \leq \text{target}$	NOT_TESTED
V-002	SAR compliance	IEEE 1528	$\text{SAR} \leq 1.6 \text{ W/kg}$	NOT_TESTED
V-003	FCC Part 15.250 compliance	FCC certification	Pass	NOT_TESTED
V-004	EMC IEC 60601-1-2	EMC test	Pass	NOT_TESTED

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VAL-001	Clinical localization accuracy	Clinical study vs imaging	Clinically useful accuracy	NOT_PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
Implant antenna	Platinum-iridium	Standard implantable electrode material	ISO 10993 + antenna characterization	CANDIDATE
Implant antenna	Gold	Biocompatible conductor	ISO 10993 + antenna characterization	CANDIDATE
Implant housing	Titanium	Standard implantable housing	ISO 10993 + hermeticity	CANDIDATE
Implant housing	Parylene-C (polymer encapsulation)	Standard implantable polymer enclosure	ISO 10993 + hermeticity + RF transparency	CANDIDATE — preferred for RF transparency

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
01	UWB transmitter IC	1	COTS_CANDIDATE	IC (COTS)	Multiple (Qorvo, etc.)	CRITICAL	Bench characterization + EMC
02	Implantable antenna (custom UWB)	1	CUSTOM_COMPONENT	OWNED — biocompatible metal (e.g., Pt)	Internal (or specialized vendor)	HIGH	Antenna measurement in phantom
03	Implant housing + battery 1	1	CUSTOM_COMPONENT	Polymer or polymer	Multiple	HIGH	Hermeticity + biocompatibility
04	External receiver array (4-16 antennas)	1 set	COTS_CANDIDATE	IC (COTS)	Multiple	HIGH	Calibration + system test

13. Manufacturing

Status: ENGINEERING_CANDIDATE — no catheter-integrated UWB process validated

Candidate Processes:

- Antenna microfabrication (lithography or laser-cut): Standard microfabrication - Custom process for catheter-integrated UWB antenna
- Hybrid circuit assembly (UWB IC + antenna + battery): Standard hybrid assembly - Miniaturization required
- Hermetic encapsulation (titanium canister or parylene): Standard implantable encapsulation - Parylene preferred for RF transparency

14. External Evidence

Source 1: Cath Lab RFID Tracking Reaps Big ROI | DAIC

URL: <https://www.dicardiology.com/article/cath-lab-rfid-tracking-reaps-big-roi>

Excerpt: Passive RFID helps instantly track thousands of items , active RFID can track patients through the perioperative process.

Source 2: RFID for Medical Devices

URL: <https://www.hidglobal.com/solutions/rfid-medical-devices>

Excerpt: Embed RFID tags into medical devices to prevent counterfeiting, leverage auto calibration, track assets, and protect patients. Used by medical OEMs.

Source 3: Localization and Tracking of Implantable Biomedical Sensors

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5375869>

Excerpt: by I Umay · 2017 · Cited by 63 — In RSS-based localization systems, the use of transmitter-receiver pairs allows transmitting a signal from the biomedical sensor to some receivers placed on the ...

Source 4: Rfid enabled medical devices and associated systems

URL: <https://patents.google.com/patent/EP4216804B1/en>

Excerpt: An RFID tag can be coupled with a medical device and can be interrogated by an RFID emitter to determine information about the medical device . This information ...

Source 5: Medical Device Management

URL: <https://identimedical.com/medical-device-management>

Excerpt: With more sensors, and improved implant tracking , see how TotalSense smart cabinet uses RFID to closely monitor inventory. TOTALSENSE RFID Smart Cabinet.

15. Transfer Boundary

YOU RECEIVE:

- + UWB localization concept + R1 correction (SAR-bounded accuracy)
- + TOA + SAR + tissue propagation governing equations
- + Bench test protocols (link budget, SAR, accuracy, EMC, FCC)
- + External precedent catalog (UWB medical radar, FCC Part 15.250, IEEE 1528)
- + Critical UNKNOWN disclosure (achievable accuracy, antenna efficiency, multipath)
- + This engineering dossier with failure analysis + build plan

YOU MUST DEVELOP:

- > Production UWB transmitter design (IC selection, modulation, power management)
- > Implantable UWB antenna design (small form factor)
- > External receiver array (production hardware + localization software)
- > Manufacturing process for catheter-integrated UWB implant
- > Regulatory submission (FDA + FCC combination; complex)
- > Clinical validation (localization accuracy vs imaging)
- > FCC certification + IEC 60601-1-2 compliance

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- Achievable localization accuracy given SAR constraint and tissue attenuation — UNKNOWN (critical)
- Optimal center frequency balancing resolution and attenuation — UNKNOWN
- Achievable antenna efficiency in 2 mm catheter form factor — UNKNOWN
- Multipath magnitude in vivo — UNKNOWN
- Clinical accuracy requirement for shunt position verification — UNKNOWN
- Battery life for active UWB transmitter — UNKNOWN
- Regulatory pathway (FCC + FDA combination) — UNKNOWN specifics

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	UWB TX + RX link behavior	UWB development kit (Delaware)	Path loss, SAR, localization accuracy	Link budget, SAR, localization accuracy	UWB link budget, SAR, localization accuracy	6 weeks
WP-02	Implantable UWB antenna, prototype	EM simulation software	Antenna efficiency, SAR	Efficiency > 10%, SAR < 1.6 W/kg	Antenna design report	2 weeks
WP-03	SAR simulation model	EM simulation software	SAR (SISIMO) 1g	SAR < 1.6 W/kg at target	SAR simulation report	8 weeks
WP-04	Tissue phantom with IR shunt positioning	UWB localization system	Localization accuracy, multipath	Localization accuracy < 10 mm, multipath < 10 dB	Localization accuracy report	6 weeks
WP-05	SAR measurement in phantom	SAR measurement system	SAR measurement	SAR < 1.6 W/kg	SAR measurement report	6 weeks

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-06	EMC test articles (UWB test plan table 6.1)	EMC test chamber and EMC test equipment	EMC emissions at 6001-1-2	EMC test results	EMC report	8 weeks (external lab)
WP-07	FCC certification test articles	FCC approved test chamber	FCC compliance measurements	FCC certification	FCC certification	12 weeks (external)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 5 governing equations and a 7-step build plan. The next experiment is clearly specified. Target buyer: Medical device localization company; neuroshunt manufacturer.

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: SAR limit (1.6 W/kg) constrains transmit power below useful SNR at >5cm tissue depth

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: SAR limit (1.6 W/kg) constrains transmit power below useful SNR at >5cm tissue depth

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.